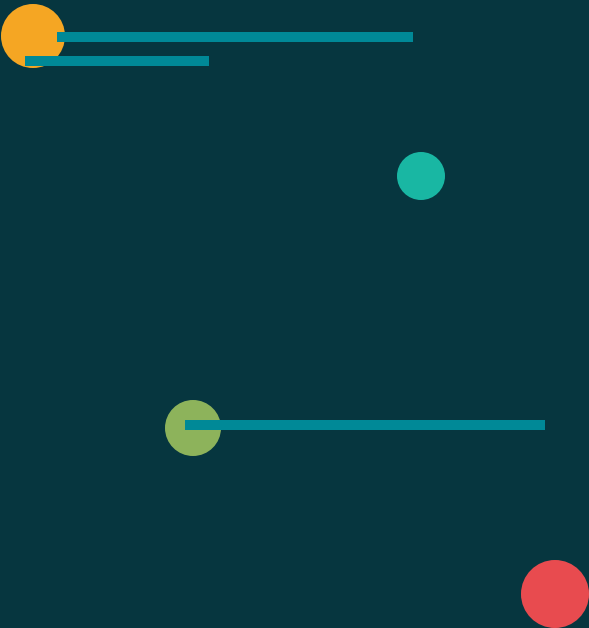


Route2Zero

Transparent corridor intelligence for a just e-jEEPney transition

MVP demonstration • Metro Manila • 14 August 2026



DATA → EVIDENCE → DECISION

Cities need a shortlist they can challenge—not a black box

Route2Zero turns incomplete transport evidence into a transparent decision conversation.

The decision

Where should cities investigate e-jEEPney conversion first?

A ranking is useful only when its sources, proxies, missing fields, and adjustable assumptions stay visible.



Climate

Surface high-activity corridors



Equity

Keep density exposure visible



Feasibility

Flag grid and operator gaps

The MVP starts with Metro Manila and ends with accountable owners

A six-month pilot moves from historic-data screening to city-validated evidence.

1,522

jeepney route-direction records
screened

3

spotlight LGUs proposed for validation

1

city-validated shortlist as the decision
output



LGU climate and transport
teams



Operators and drivers



Riders and communities



Utilities and financing
partners

One screen connects the network view to a defensible route story

Filters change the lens; every route keeps its score components and confidence labels.

Route2Zero
Decision support for a just e-jeepney transition

City / LGU lens

☐ Include unverified equity routes

Live score weights

Emissions potential
35

Equity density proxy
35

Grid feasibility
15

Operator readiness
15

Weights are normalized automatically. AI never changes these numbers.

METRO MANILA • ROUTE ELECTRIFICATION INTELLIGENCE

Route2Zero

Rank e-jeepney corridors by climate impact, equitable service, grid context, and operator readiness — with every proxy and placeholder visible before anyone asks.

Data confidence is part of the product. GTFS service dates span 2013–2020; 1,520 of 1,522 geometries connect ordered stops rather than road-snapped shapes. Emissions are an activity proxy, equity uses 2020 WorldPop density rather than settlement boundaries, grid is one Luzon baseline, and operator readiness defaults to 50 pending cooperative workshops.

Routes in view 1,522	Complete scores 1,521	Top priority score 71.4	Equity source ⓘ WorldPop ...
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Explore priority corridors

Select a line on the map or use the route finder. Color moves from red (lower priority) through amber to green (higher priority); lighter lines are stop-sequence approximations.

1

Choose a city lens

2

Adjust four weights

3

Explore the network

4

Open a route explanation

The audit is strict about what the historic GTFS can—and cannot—prove

The canonical Sakay master feed is reproducible, but its calendar ends in 2020.

1,717

routes in the canonical feed

1,522

PUJ route-direction records

0

broken foreign-key references

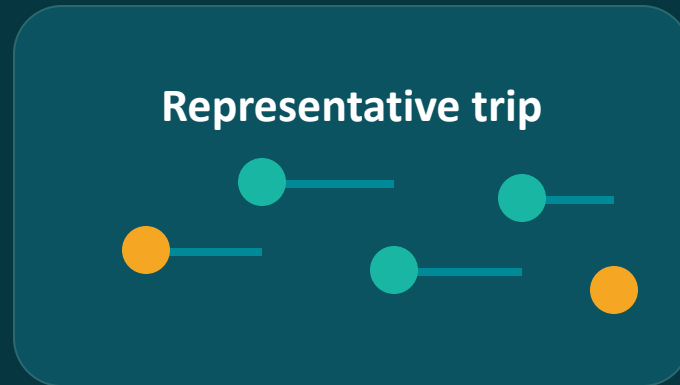
Historic-data boundary

Service dates span 2013–2020. Route2Zero is a screening baseline, not a claim about current operations.

Geometry confidence travels with every route line

Only two jeepney records have GTFS shapes; the fallback connects ordered stops and labels the approximation.

2
GTFS shape



1,520
stop-sequence
approximations

Why it matters: route order is useful for screening, but the corridor must be ground-checked before investment.

Four interpretable signals combine into one adjustable priority score

The default weights are visible and normalized whenever a user changes them.



Emissions potential rewards service activity—not claimed fuel savings

Length × estimated daily trips is a transparent activity proxy.

98.2

Top route activity score

length (km)

×

trips/day

→

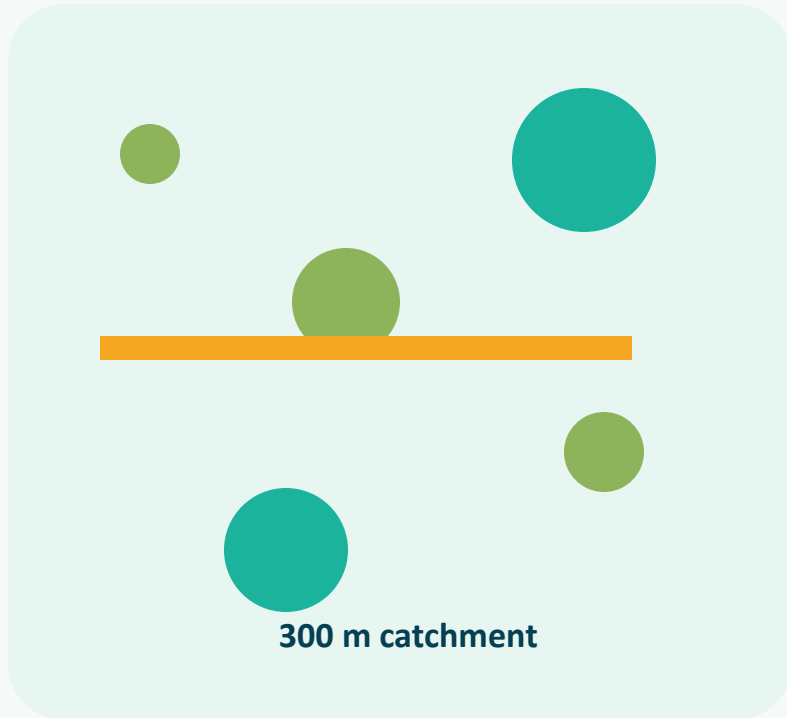
percentile

Interpretation

Higher values identify corridors with more transport activity to examine. They do not estimate fleet fuel use, grid losses, or verified CO₂ reductions.

WorldPop shows density exposure, not informal-settlement status

A 300-metre route catchment is scored against the densest quarter of positive NCR raster cells.



77.4

Top route density-exposure score

Safe claim

The corridor overlaps high-density population cells.

Unsafe claim

The corridor serves an informal settlement.

The current grid signal is one Luzon baseline—not local charger capacity

It is intentionally constant across routes until depot, feeder, tariff, and resiliency evidence arrives.

16.1%

2024 Luzon renewable share

14,550 GWh

renewable generation

÷

90,269 GWh

total generation

Required before investment

- candidate depot
- distribution feeder
- interconnection window
- tariff and demand charges
- solar / storage option

Operator readiness starts neutral and is designed to be replaced

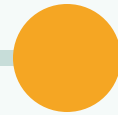
Every route receives 50/100 until a cooperative workshop produces documented evidence.

50

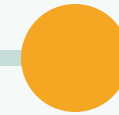
neutral placeholder



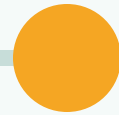
Fleet &
depot



Finance



Operations



Governance

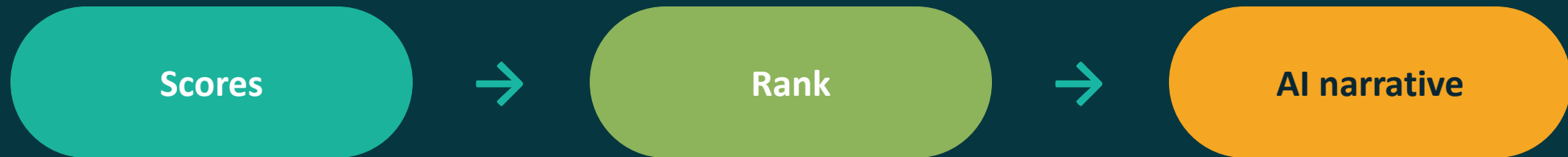
Override workflow

Route-level evidence can replace 50 while retaining notes, source, date, and reviewer.

The formula is public; AI only explains the result after ranking

Deterministic scores and rankings remain identical whether the API is available or not.

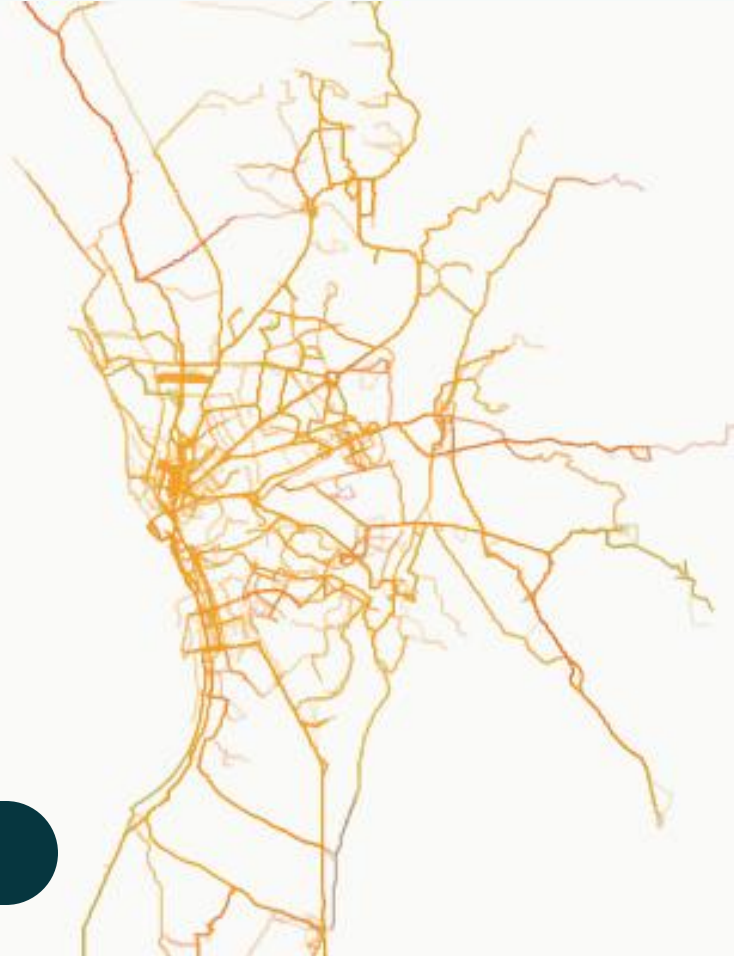
$$0.35E + 0.35Q + 0.15G + 0.15O$$



Never score. Never reorder. Never fill missing evidence silently.

The map shows priority and geometry confidence together

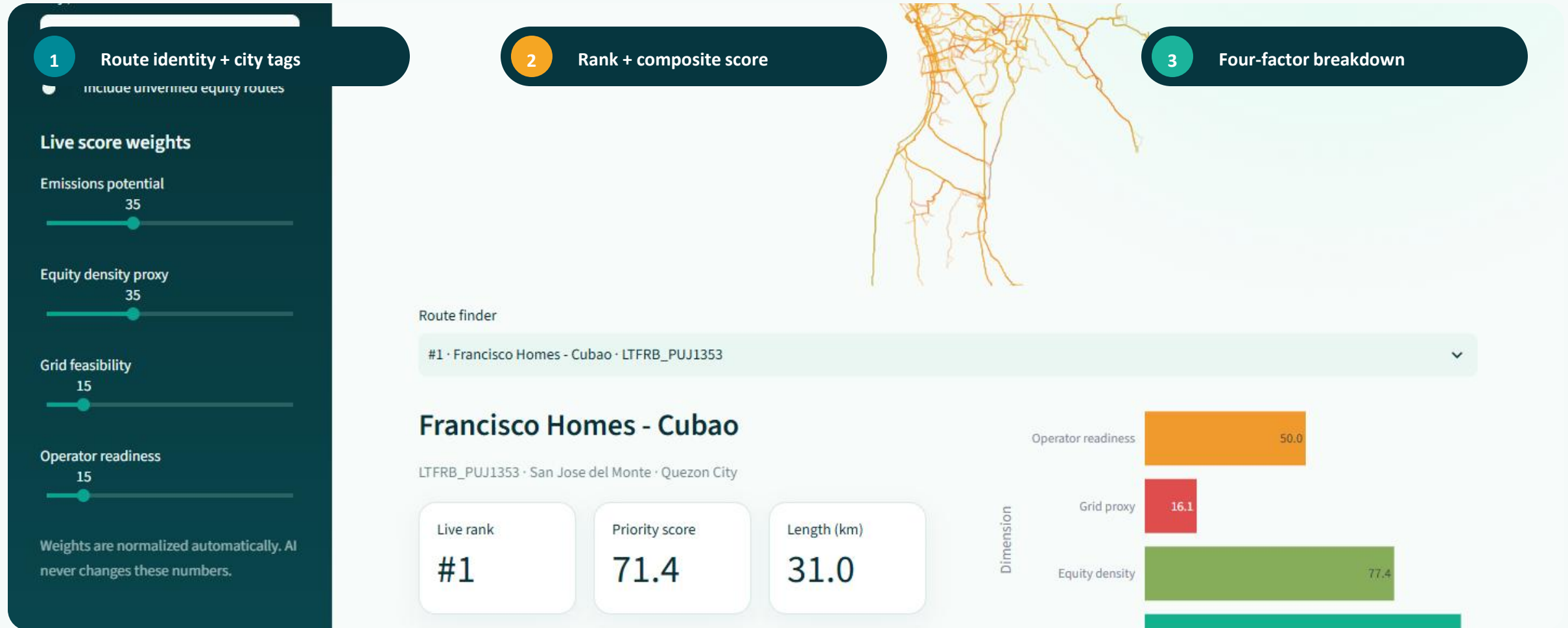
Color moves red → amber → green; lighter lines indicate stop-sequence approximation.



CLICK A LINE → OPEN ROUTE EVIDENCE

A selected corridor exposes score, rank, components, and explanation

The #1 screen also shows exactly where evidence is strong and where it is shared or missing.



A city lens changes the shortlist without hiding the shared assumptions

Marikina returns 53 routes; the leading filtered result scores 60.6.

53

routes tagged to Marikina

60.6

top filtered score

Same weights. Same provenance. New local lens.

Route2Zero

Decision support for a just e-jeepney transition

City / LGU lens

☐ Include unverified equity routes

Live score weights

Emissions potential



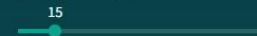
Equity density proxy



Grid feasibility



Operator readiness



Weights are normalized automatically. AI never changes these numbers.

Route finder

#41 - Recto - SSS Village via Aurora Blvd., Espana - LTFRB_PUJ1577

Recto - SSS Village via Aurora Blvd., Espana

LTFRB_PUJ1577 - Marikina - Manila

Live rank

#41

Priority score

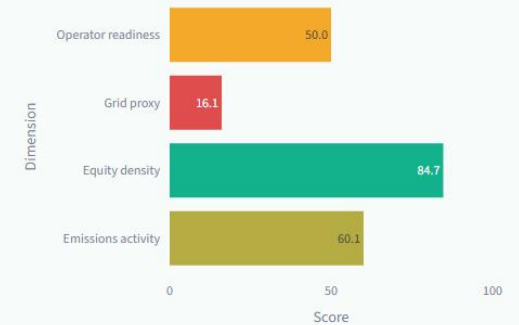
60.6

Length (km)

19.1

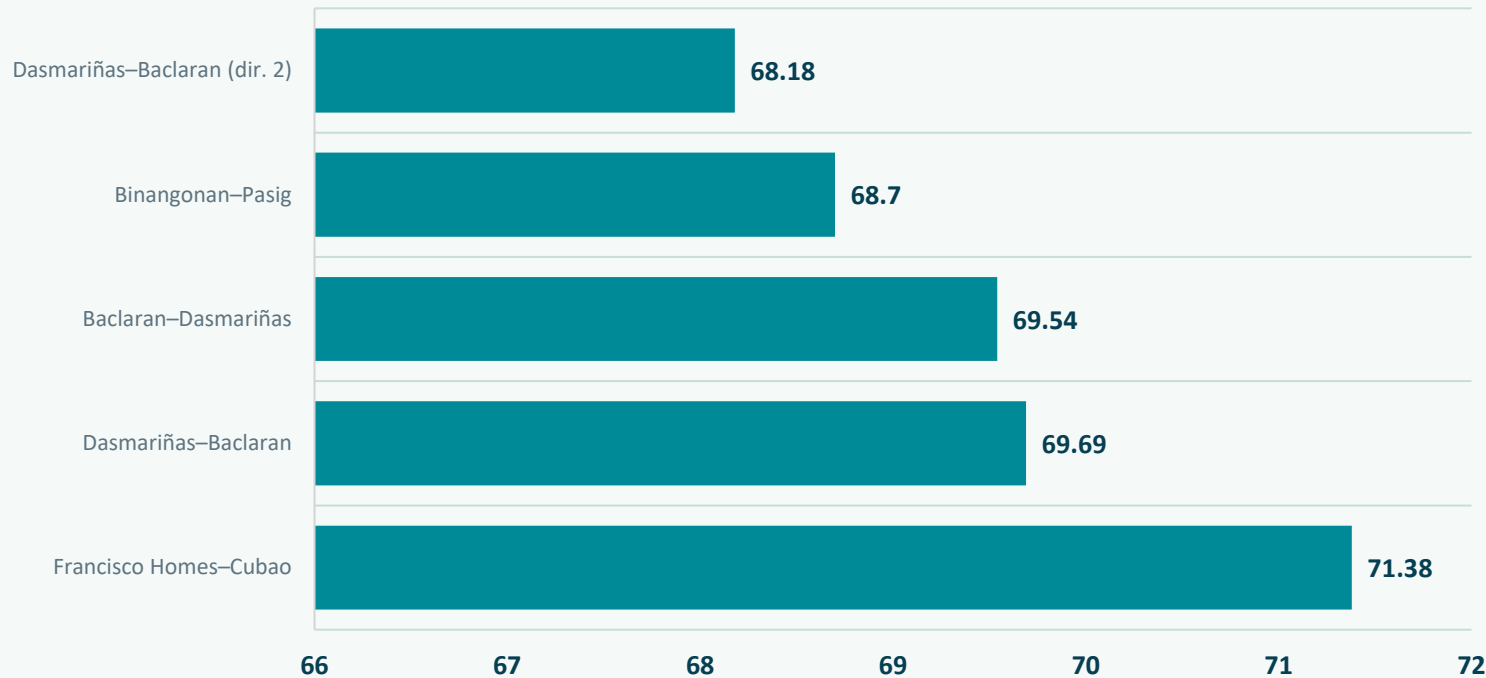
Deterministic explanation • offline-safe

This route ranks here because its strongest signal is high-density corridor exposure (85/100); the 16/100 grid value is a shared Luzon proxy and the 50/100 operator value remains a neutral placeholder



The leaderboard creates a validation queue—not a funding decision

Top five route-direction records under the default 35 / 35 / 15 / 15 weights.



Validation queue

- 1 Refresh route
- 2 Confirm operator
- 3 Screen depot
- 4 Ask riders
- 5 Re-run scenarios

Every shortlisted route can leave the dashboard as a policy-ready brief

The brief packages the score, source boundaries, uncertainties, and next evidence requests.

ROUTE2ZERO POLICY BRIEF

Francisco Homes–Cubao

Priority score 71.4 · Live rank #1

WHY IT SURFACES

High activity and density exposure

WHAT REMAINS UNKNOWN

Current service · operator evidence · depot capacity

8

briefs generated

CSV

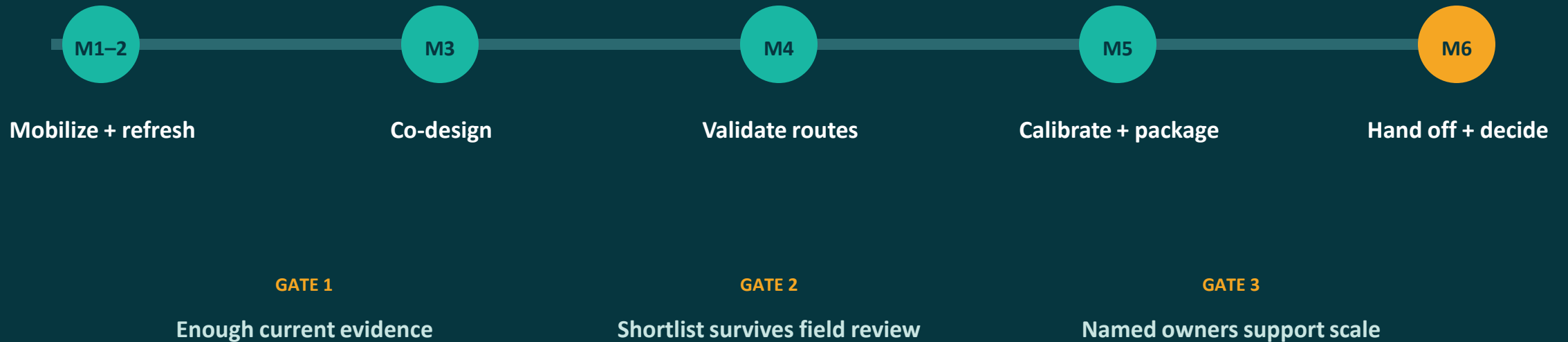
rank export

JSON

AI cache

Six months convert a screening model into a city-owned decision process

Three decision gates stop the pilot from scaling on weak evidence.



Eight leads keep the pilot accountable

Product, evidence, engagement, and assurance stay cross-functional; every artifact retains one owner.



John Marwin Ebona

Project Lead · Policy



Joaquin Sarmiento

Transport Analytics



Andrei Dela Cruz

Climate · Grid



Tj Moreno

Operator Partnerships



Prince Marl

Data Engineering



Isaac Marcus

Product · Streamlit



Russel Mendez

Equity · Community



JM Palaganas

QA · Docs · Demo

Weekly delivery review · Biweekly partner clinic · Monthly decision gate



THE ASK

Choose three spotlight LGUs and open the evidence refresh

Route2Zero is ready to screen. The next step is to validate routes, operators, communities, and charging conditions together.

01

Name city owners

02

Confirm data access

03

Schedule the first clinic